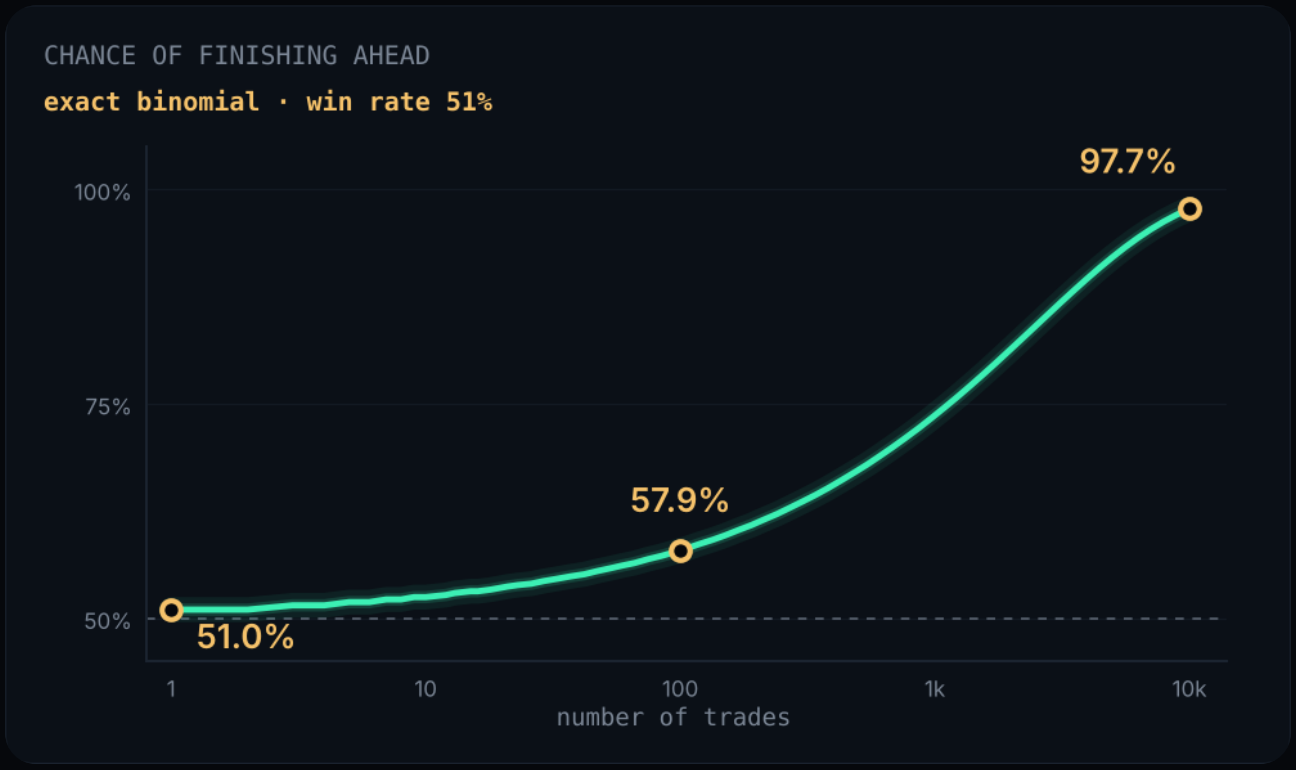
ZERO TO QUANT TRADER

EP. 01 · BEGINNER · FREE GUIDE

What a Quant Does

a tiny edge, repeated



- ✓ all 100 episodes, module by module
- ✓ what to study, in what order
- ✓ tools, books and a 12-week plan

simulated data · educational only · not financial advice · code is for learning, not production trading

What this guide covers

The reel showed one idea: a quant does not predict the market, a quant finds a small edge and repeats it until the noise averages out. This guide turns that idea into a path. It explains the arithmetic behind the reel, then lays out all 100 episodes of the series in order, what you need to know before each module, and a 12-week study plan you can follow alongside the reels.

Every number in the first two sections comes from the code in this guide.

1. The idea behind the reel

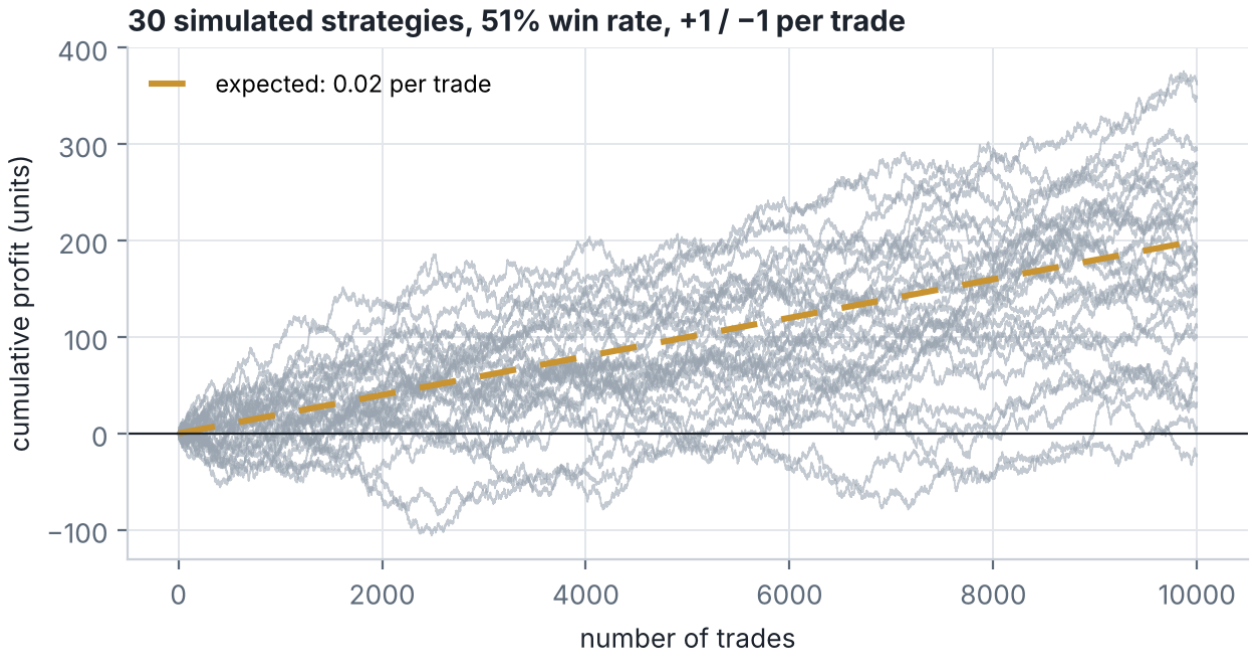
Take a strategy that wins 51% of its trades. Each trade pays +1 unit or −1 unit. The average gain per trade is the *edge*:

$$\text{edge} = p \cdot (+1) + (1-p) \cdot (-1) = 2p - 1 = 0.02$$

Two cents per trade is small. A single trade is almost a coin flip: you finish ahead with probability 51.0%. The point is what repetition does. After n trades the total edge is $0.02\,n$, while the noise (the standard deviation of the total) grows only as $\sqrt{n}$:

$$\text{expected profit} = 0.02\,n, \qquad \text{noise} \approx \sqrt{n}, \qquad \frac{\text{edge}}{\text{noise}} = 0.02\sqrt{n}$$

At $n = 10{,}000$ the expected profit is 200 units and the noise is about 100, so the ratio is 2. At $n = 100$ the ratio is only 0.2, and luck dominates.



Thirty simulated strategies with a 51% win rate. The dashed line is the expected path.

In this simulation 29 of the 30 paths finish above zero after 10,000 trades, with a median final profit of 177 against an expected 200. Yet the worst path dipped to −106 along the way. An edge does not remove the bad stretches, it outlasts them.

2. The code

The function below gives the chance of finishing ahead after n trades. A tie (finishing exactly level) counts as half, which keeps the curve smooth.

```
from scipy.stats import binom

def ahead(n, p):
    # P(strictly ahead) + half of P(level)
    return binom.sf(n / 2, n, p) + 0.5 * binom.pmf(n / 2, n, p)

for n in (1, 10, 100, 1_000, 10_000):
    print(f"{n:>6} trades: {ahead(n, 0.51):.1%} ahead")
```

trades	chance of finishing ahead
1	51.0%
10	52.5%
100	57.9%
1,000	73.6%
10,000	97.7%

The curve rises slowly at first and then quickly: the ratio $0.02\sqrt{n}$ needs to reach about 2 before the outcome becomes reliable.

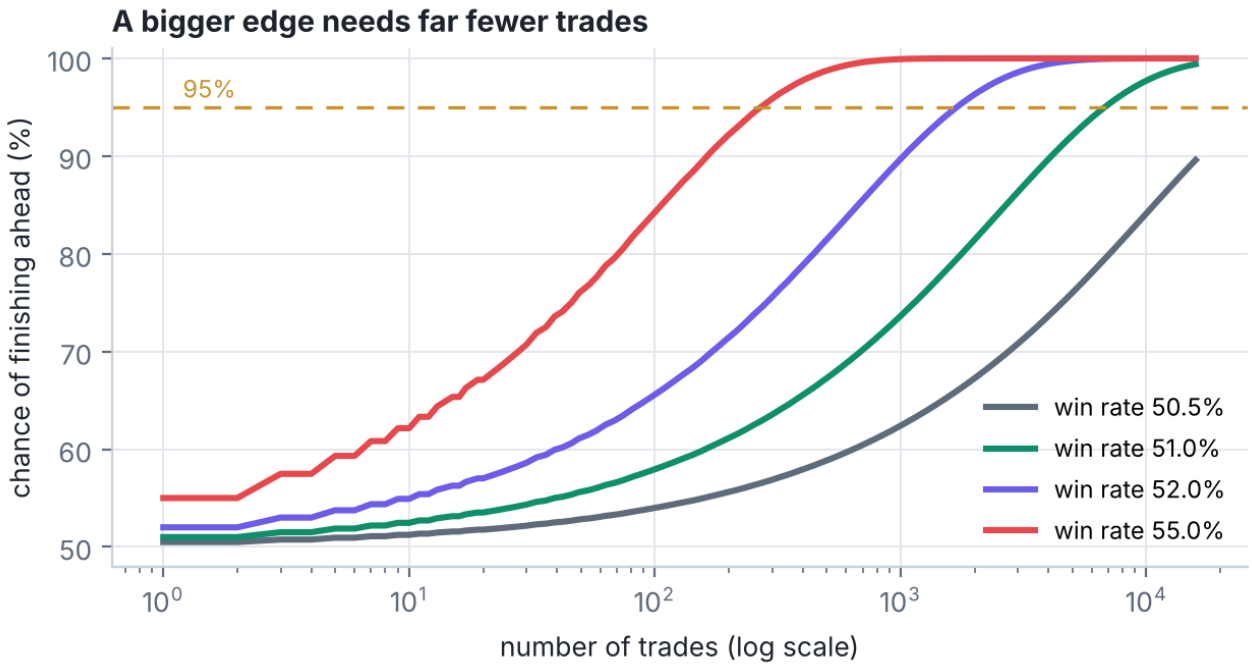
3. How many trades is enough?

It depends almost entirely on the size of the edge. The table gives the smallest number of trades for a 95% chance of finishing ahead.

win rate	edge per trade	trades needed
50.5%	0.010	27,055
51.0%	0.020	6,763
52.0%	0.040	1,691
55.0%	0.100	269
60.0%	0.200	67

Halving the edge multiplies the trades needed by four, because of the square root. This is why real quant work is split in two: finding an edge, and doing it cheaply and often enough for the maths to

work. It is also why costs matter so much (Module 1) and why an edge that is only a statistical accident disappears when you test it properly (Modules 2 and 6).



Chance of finishing ahead against number of trades, for four win rates.

Key idea. A quant does not need to be right often. A quant needs a positive edge, enough repetitions, and risk small enough to survive the bad stretches while the repetitions accumulate.

4. What this simple model leaves out

- **Costs.** Every trade pays spread and fees. A 0.02 edge can be erased by a 0.02 cost.
- **Payoffs are not ±1.** Real wins and losses differ in size. Expected value, not win rate, decides (Episode 11).
- **Edges decay.** Others find the same pattern and it shrinks.
- **Trades are not independent.** Correlated positions behave like fewer trades than you counted.
- **Sizing.** Even a true edge can ruin you if each bet is too large (Episodes 39 and 40).

5. The 100-episode roadmap

Episodes run daily from 6 November 2026 to 13 February 2027. Each one ends with a comment keyword; comment it on that episode and the free material arrives in your DMs.

Module 1 — Market Basics (Ep. 1–10)

How markets actually trade: order books, spreads, costs, who is on the other side.

Ep.	Title	Free with the comment keyword
1	What a quant actually does	**ROADMAP**: the 100-episode roadmap and study plan (PDF)
2	The order book in 30 seconds	**BOOK**: order-book cheat sheet (PDF)
3	The spread you pay without noticing	**SPREAD**: trading-cost calculator (Google Sheet)
4	Market vs limit orders	**ORDERS**: order-types one-pager (PDF)
5	Who's on the other side of your trade	**WHO**: adverse selection explained (PDF)
6	Auctions, ticks and halts	**AUCTION**: market-structure guide (PDF)
7	The quant's toolbox	**TOOLBOX**: instruments cheat sheet with ES, NQ and MNQ specs (PDF)
8	Leverage: one micro contract, $60k of Nasdaq	**LEVERAGE**: leverage and margin calculator (Google Sheet)
9	Simple vs log returns	**RETURNS**: returns and compounding notebook
10	Costs: the silent strategy killer	**COSTS**: transaction-cost checklist (PDF)

Module 2 — Probability & Statistics (Ep. 11–20)

Expected value, variance, distributions and the classic interview puzzles, taught through trades.

Ep.	Title	Free with the comment keyword
11	Expected value: the only number that matters	**EDGE**: win rate and payoff calculator (Google Sheet)
12	Volatility drag	**DRAG**: volatility-drag guide with simulation code (PDF)
13	The normal distribution lies about crashes	**TAILS**: fat-tails notebook, normal vs Student-t
14	Bayes: update like a quant	**BAYES**: Bayes cheat sheet and 10 interview puzzles (PDF)
15	Correlation breaks when you need it	**CORR**: rolling-correlation notebook
16	Why the casino always wins	**CASINO**: law of large numbers and trade count (PDF)
17	Is your backtest just luck?	**TSTAT**: significance-test notebook
18	Regression: alpha vs beta	**ALPHA**: alpha and beta regression notebook
19	Test 100 strategies, find 5 fake winners	**LUCK**: multiple-testing guide: Bonferroni, deflated Sharpe (PDF)
20	Monte Carlo: 10,000 alternate histories	**SIM**: equity-curve resampling notebook

Module 3 — Python for Quants (Ep. 21–28)

A free, working Python setup, then data, pandas and plotting for finance.

Ep.	Title	Free with the comment keyword
21	A quant setup for $0	**SETUP**: setup guide and requirements file (PDF)
22	20 years of prices in 3 lines	**DATA**: price downloader script
23	pandas for traders	**PANDAS**: pandas cheat sheet (PDF)
24	Survivorship bias	**SURVIVOR**: point-in-time universe builder (code)
25	Plot like a hedge fund	**PLOT**: dark-theme plotting template (same look as the reels)
26	Vectorize or die	**FAST**: numpy vectorization notebook
27	Your first backtest in 20 lines	**BACKTEST**: 20-line backtest with notes (code + PDF)
28	Look-ahead bias	**LOOKAHEAD**: 10 backtest bugs checklist (PDF)

Module 4 — Time Series (Ep. 29–38)

Returns, stationarity, autocorrelation, ARMA and GARCH on real price series.

Ep.	Title	Free with the comment keyword
29	Can you beat a coin flip?	**RANDOM**: random-walk generator and "spot the real chart" quiz
30	Stationarity	**STATIONARY**: ADF test notebook
31	Does yesterday predict today?	**ACF**: autocorrelation notebook
32	Mean reversion vs momentum	**REGIME**: Hurst exponent code and guide (PDF)
33	ARIMA in 30 seconds	**ARIMA**: ARIMA notebook
34	Volatility clustering	**CLUSTER**: volatility-clustering chart code (PDF)
35	GARCH: forecasting volatility	**GARCH**: GARCH(1,1) notebook
36	Cointegration	**COINT**: Engle-Granger notebook
37	Pairs trading, tested over 20 years	**PAIRS**: full pairs-trading backtest code
38	Kalman filter	**KALMAN**: adaptive hedge-ratio notebook

Module 5 — Portfolio & Risk (Ep. 39–50)

Mean-variance, Kelly, risk measures and how to size a bet.

Ep.	Title	Free with the comment keyword
39	Kelly criterion: how much to bet	**KELLY**: Kelly sizing calculator and guide (Google Sheet + PDF)
40	Why pros bet half-Kelly	**HALF**: full vs half Kelly drawdown simulation (notebook)
41	Sharpe, Sortino, Calmar	**SHARPE**: performance-metrics calculator (Google Sheet + Python)
42	The drawdown that ends careers	**DRAWDOWN**: drawdown analyzer code
43	VaR and expected shortfall	**VAR**: VaR and expected shortfall notebook
44	Markowitz: diversification maths	**FRONTIER**: efficient-frontier notebook
45	Why optimizers blow up	**OPTIMIZER**: shrinkage and robust weights guide (PDF + code)
46	Risk parity	**PARITY**: risk-parity weights calculator
47	Volatility targeting	**VOLTARGET**: volatility-targeting code
48	A quant book's portfolios, out of sample	**LAZY**: 25 portfolios backtested (PDF)
49	Why All Weather broke in 2022	**WEATHER**: 2022 post-mortem (PDF)
50	Three engines: survived 2008 and 2022	**ENGINES**: full recipe and backtest code, plus the Ep. 1–50 handbook

Module 6 — Strategies & Backtesting (Ep. 51–64)

Momentum, mean reversion, pairs; how to backtest without fooling yourself.

Ep.	Title	Free with the comment keyword
51	Time-series momentum	**TREND**: time-series momentum code and paper summary
52	The Turtle experiment	**TURTLE**: the original Turtle rules in Python
53	Buy last year's best stock	**WINNER**: full backtest with year-by-year picks
54	Momentum 12-6-3 on the Nasdaq-100	**MOMENTUM**: strategy code and robustness grid
55	Dual momentum	**DUAL**: dual-momentum code
56	Sector momentum, since 1927	**SECTOR**: sector-momentum notebook on Fama-French data
57	Betting Against Beta	**BETA**: Betting Against Beta replication code
58	Combining weak strategies	**COMBO**: equal-risk combination code
59	The last 30 minutes	**CLOSE**: intraday-momentum test code and verdict (PDF)
60	The Noise Area strategy	**NOISE**: full strategy rules and code
61	VWAP: the "holy grail"?	**VWAP**: VWAP strategy code and paper notes
62	Overfitting	**OVERFIT**: overfitting checklist (PDF)
63	Walk-forward testing	**WALK**: walk-forward framework code
64	The maths of a prop-firm challenge	**PROP**: expected-value calculator per firm (Google Sheet)

Module 7 — Stochastic Calculus (Ep. 65–70)

Brownian motion, Ito's lemma and the maths under option pricing.

Ep.	Title	Free with the comment keyword
65	Brownian motion	**BROWNIAN**: Brownian-motion simulator (notebook)
66	Bachelier, 70 years early	**BACHELIER**: the history plus Bachelier vs Black-Scholes code (PDF)
67	Geometric Brownian motion	**GBM**: GBM path simulator
68	Itô's lemma	**ITO**: step-by-step Itô derivation (PDF)
69	Monte Carlo pricing	**MONTE**: Monte Carlo option pricer (notebook)
70	Risk-neutral pricing	**NEUTRAL**: risk-neutral pricing explained (PDF)

Module 8 — Options & Derivatives (Ep. 71–82)

Black–Scholes, the Greeks, binomial trees and hedging.

Ep.	Title	Free with the comment keyword
71	Options in 30 seconds	**OPTIONS**: payoff-diagram builder
72	Put-call parity	**ARB**: parity arbitrage sheet (Google Sheet)
73	Binomial trees	**TREE**: binomial pricer (notebook)
74	Black-Scholes from scratch	**PRICE**: Black-Scholes calculator and derivation (PDF + code)
75	Delta hedging	**DELTA**: delta-hedging simulator
76	The Greeks	**GREEKS**: Greeks cheat sheet and calculator
77	Gamma: the Greek that moves markets	**GAMMA**: dealer gamma explained (PDF)
78	GameStop 2021: a gamma squeeze	**SQUEEZE**: gamma-squeeze feedback model (PDF)
79	Theta: paid to wait	**THETA**: time-decay chart code
80	The wheel strategy, quantified	**WHEEL**: wheel rules and paper findings (PDF)
81	Futures and the basis	**CARRY**: futures fair-value calculator
82	Exotic options	**EXOTIC**: Monte Carlo pricer for barriers, digitals and Asians

Module 9 — Volatility (Ep. 83–89)

Implied vol, skew, local and stochastic volatility, including Heston.

Ep.	Title	Free with the comment keyword
83	Implied vs realised volatility	**VRP**: implied vs realised notebook
84	The volatility smile	**SMILE**: smile and skew guide (PDF)
85	The VIX	**VIX**: how the VIX is calculated, step by step (PDF)
86	Volmageddon, 5 February 2018	**XIV**: post-mortem and VIX futures term structure (PDF)
87	Harvesting the volatility premium	**VOLEDGE**: the paper's rules in code
88	Heston: volatility with a life of its own	**HESTON**: derivation, commented code, semi-analytic pricing and calibration (PDF)
89	The volatility surface	**SURFACE**: volatility-surface builder (notebook)

Module 10 — Machine Learning (Ep. 90–95)

Features, validation and models that survive out-of-sample.

Ep.	Title	Free with the comment keyword
90	ML in trading: what actually works	**BOT**: map of ML methods that hold up in markets (PDF)
91	Feature engineering	**FEATURES**: 30 price and volume features (notebook)
92	Random forests and XGBoost	**XGB**: classifier with walk-forward validation (notebook)
93	LSTMs and transformers: hype check	**LSTM**: LSTM notebook and reality check
94	Reinforcement-learning traders	**RL**: RL trading environment (code)
95	LLMs that invent alpha	**LLM**: prompt pack and factor-testing code

Module 11 — Execution & Going Live (Ep. 96–100)

Transaction costs, execution algorithms and running a strategy live.

Ep.	Title	Free with the comment keyword
96	Market impact: the square-root law	**IMPACT**: market-impact calculator (Google Sheet)
97	Optimal execution	**EXECUTE**: Almgren-Chriss notebook
98	How market makers make money	**MM**: market-making simulator
99	Knight Capital: $440M in 45 minutes	**KILL**: go-live risk checklist and kill-switch template (PDF)
100	Zero to Quant Trader: the whole map	**QUANT**: the full handbook: all 100 guides plus the roadmap

6. A 12-week study plan

The reels teach one idea each. To make them stick, work through them in modules and do one small exercise per episode. A realistic pace is 5 to 6 hours a week.

weeks	modules	what to do each week
1	1 Market Basics	Watch Ep. 1–10. Write down one number from each episode and where it came from.
2–3	2 Probability & Statistics	Redo each expected-value example by hand, then in Python. Solve the interview puzzle of each episode before the answer.
4	3 Python for Quants	Set up the environment, download prices, plot returns. Keep everything in one notebook.
5–6	4 Time Series	Run each test on two different assets. Note when the result changes.
7	5 Portfolio & Risk	Build a two-asset portfolio, compute its risk three ways.
8–9	6 Strategies & Backtesting	Backtest one simple rule with costs. Then try to break it with a different period.
10	7 and 8	Price a call by simulation and by formula; compare.
11	9 Volatility	Compute realised volatility and compare it with a published implied volatility.
12	10 and 11	Fit one simple model with a proper train/test split. Write down what execution would cost.

7. What you need before you start

- **Maths.** School algebra, basic probability, and the idea of a derivative and an integral. Everything else is built up in the episodes.
- **Programming.** None for Modules 1–2. Module 3 starts from an empty notebook.
- **Tools.** A laptop, Python 3 with NumPy, pandas, SciPy and matplotlib, and a Jupyter notebook. All free.
- **Data.** Free daily prices are enough for almost the whole series. Simulated data is always labelled as such.

8. Exercises

1. A strategy wins 55% of its trades, each paying ±1. What is the chance of finishing ahead after 100 trades? Use `ahead(100, 0.55)`.
2. A strategy wins exactly 50% of its trades. What is the chance of finishing ahead after 10,000 trades? What does that say about repetition without an edge?
3. Using `trades_needed` (code in the solutions), find how many trades give a 99% chance of finishing ahead at a 52% win rate.

9. Solutions

1. **84.1%.** A 55% win rate has edge 0.10, so after 100 trades edge over noise is $0.10 \times \sqrt{100} = 1$.
2. **50.0%.** With no edge, repetition changes nothing: the average gain is zero at every n and the spread of outcomes only widens.
3. **3,381 trades**, about twice the 1,691 needed for 95%. The function is below.

```python
def trades_needed(p, target=0.95):
    # doubling, then bisection
    hi = 1
    while ahead(hi, p) < target:
        hi *= 2
    lo = hi // 2
    while hi - lo > 1:
        mid = (lo + hi) // 2
        lo, hi = (mid, hi) if ahead(mid, p) < target else (lo, mid)
    return hi
```

10. Books and references

- Hull, J. *Options, Futures, and Other Derivatives* (Pearson).
- Shreve, S. *Stochastic Calculus for Finance II: Continuous-Time Models* (Springer, 2004).
- Grinold, R. and Kahn, R. *Active Portfolio Management* (McGraw-Hill, 2nd ed., 1999).
- Carver, R. *Systematic Trading* (Harriman House, 2015).
- Harris, L. *Trading and Exchanges: Market Microstructure for Practitioners* (Oxford University Press, 2003).
- Jansen, S. *Machine Learning for Algorithmic Trading* (Packt, 2nd ed., 2020).
- López de Prado, M. *Advances in Financial Machine Learning* (Wiley, 2018).
- Crack, T. *Heard on the Street: Quantitative Questions from Wall Street Job Interviews* (self-published).

Education only, not financial advice. All data in this guide is simulated.